A Clinical Informatics Framework for Myeloid Oncology: Scalable AI and LLM Integration for Adaptive Trial Management and Automated FDA Compliance

Senthilkumar Vijayakumar[1], Shaunak Pai Kane[2], Selvavaani Senthilkumar[3], Dr. Parameshwari Vaiayapuri MBBS[4], Filious Louis[5]

IEEE Senior Member[1,3,5] & IEEE Member[2]

{[1]senthil.vijayakumar, [2]shaunak.paikane, [3]selvavaani, [5]louisfi}@ieee.org

***Abstract*— The advancement of treatment strategies in myeloid malignancies increasingly depends on the convergence of precision oncology, clinical informatics, and regulatory innovation. These disorders, including acute myeloid leukemia and related conditions, are driven by diverse genetic alterations, cytogenetic abnormalities, and hematologic parameters. Such complexity necessitates individualized therapeutic planning and adaptive trial designs. Clinical studies in this space now face layered challenge balancing molecularly defined eligibility criteria with protocol-driven oversight, predictive monitoring, and the growing demand for accelerated regulatory submissions [1], [2].**

To address these challenges, this research introduces an artificial intelligence–enabled clinical trial framework designed for treatment studies in myeloid cancers. The framework integrates electronic data capture platforms such as Medidata Rave and Patient Cloud with real-time clinical rule interpretation, machine learning–driven risk modeling, and automated regulatory document generation. Key clinical variables including patient performance status, bone marrow blast percentage, peripheral blood counts, and biomarker profiles such as FLT3, IDH1, and NPM1 are continuously tracked and evaluated against protocol-defined thresholds. Supervised learning algorithms are employed to assess eligibility dynamically, forecast early adverse event risk, and detect deviation trends before they escalate into protocol violations [3], [4].

Central to the system is a regulatory automation module that leverages large language models trained on the structure of FDA submissions. This module generates compliance-ready clinical summaries, SDTM-aligned datasets, and trial master file artifacts in accordance with established regulatory standards [3]. Results from simulation-based benchmarking show notable improvements, including higher compliance rates, better predictive accuracy for patient outcomes, and substantial reductions in cycle times for generating clinical documentation. Taken together, this study outlines a scalable framework for intelligent, regulatory-ready, AI-supported clinical trials in precision hematology. By combining molecular stratification with real-time informatics and regulatory automation, the framework offers a pathway to accelerate therapeutic development in myeloid malignancies while preserving clinical rigor and integrity.

Keywords*— *Precision Oncology, Myeloid Malignancies, Acute Myeloid Leukemia, Adaptive Clinical Trials, Machine Learning, Electronic Data Capture (EDC), Regulatory Automation, Large Language Models (LLMs).

I. Introduction

Myeloid malignancies comprise a diverse group of hematologic disorders defined by abnormal proliferation, impaired differentiation, and the inherent risk of progression toward leukemic transformation. Within this spectrum, acute myeloid leukemia (AML), myelodysplastic syndromes (MDS), and chronic myelomonocytic leukemia (CMML) present unique molecular and clinical complexities. The course of these diseases is dictated by a combination of somatic mutations, cytogenetic abnormalities, and baseline hematologic status, all of which influence therapeutic response and prognosis. Consequently, clinical investigations in this field require trial designs that allow for dynamic patient stratification and flexible eligibility criteria that reflect evolving scientific and clinical insights [5].

A cornerstone of this stratification is biomarker profiling, which enables precise disease classification and informs therapeutic allocation. In practice, bone marrow and peripheral blood samples are analyzed to detect molecular drivers that shape disease biology and therapeutic response. Biomarkers defined as measurable biological indicators of normal or pathological processes serve as actionable determinants of treatment strategy. For example, recurrent mutations in FLT3, NPM1, IDH1, and IDH2, coupled with cytogenetic profiles and hematologic indices, guided both the assignment of patients to investigational arms and the continuation of standard-of-care approaches. In biomarker-driven frameworks such as MyeloMATCH, patients with disease-defining molecular features may be directed toward precision treatment arms, whereas those without qualifying markers may remain in conventional pathways under structured Tier Advancement models [5]. This approach enhances therapeutic alignment with disease biology, thereby maximizing efficacy and minimizing avoidable toxicity.

The shift toward biomarker-informed and targeted protocols has fundamentally reshaped the operational and regulatory demands of clinical trials. Traditional

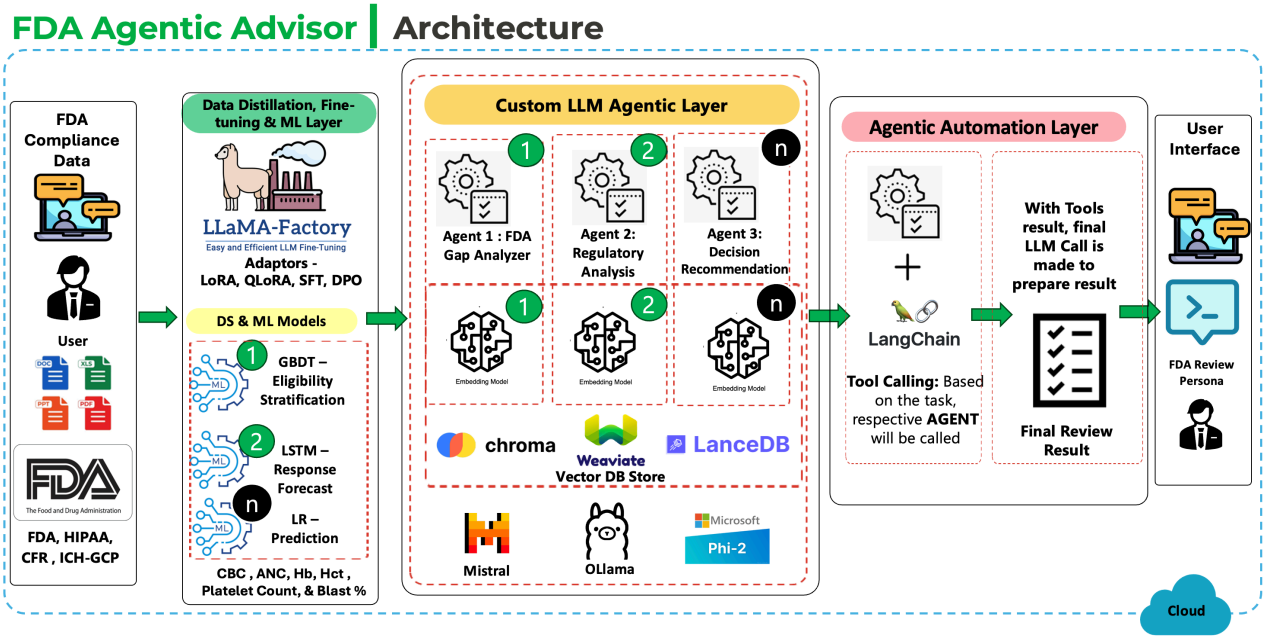

Fig. 1. System Design & Architecture of the FDA Agentic Advisor: An LLM-Driven Clinical Informatics Framework for Adaptive Trial Management and Automated Regulatory Compliance in Myeloid Oncology

mechanisms static case report forms, delayed site monitoring, and reactive handling of protocol deviations often fail to capture the speed and granularity needed for modern oncology studies. Inconsistent site performance and variable patient adherence further complicate trial integrity, creating challenges in ensuring safety, fidelity, and regulatory compliance.

In response, electronic data capture platforms such as Medidata Rave and Patient Cloud have introduced standardized frameworks for centralized data acquisition, remote monitoring, and integration of patient-reported outcomes. Yet, their transformative potential lies in combining these platforms with intelligent computational systems. Machine learning models, trained on clinical and biomarker datasets, can provide real-time eligibility verification, early risk detection, and adaptive cohort assignment. More importantly, these models can be linked directly with regulatory processes to produce structured submissions aligned with the expectations of oversight bodies such as the United States Food and Drug Administration (FDA) [6], [7], [8].

This study proposes a clinical trial framework tailored for myeloid malignancies that integrate artificial intelligence with regulatory automation. The framework leverages structured datasets from Medidata Rave, applies supervised learning techniques anchored in protocol-specific thresholds, and utilizes large language models to generate compliance-ready documentation. These outputs are designed in accordance with FDA guidance on AI/ML-enabled systems and predetermined change control plans. By aligning computational precision with clinical expertise, the system aims to strengthen the scientific rigor, operational efficiency, and regulatory transparency of treatment studies in hematologic oncology.

II. System Design and Architecture

The transformation of clinical trial infrastructure in hematologic oncology particularly in the context of myeloid malignancies requires the seamless integration of molecular insights, real-time data systems, predictive modeling, and regulatory readiness. A prime example is the Master Screening and Reassessment Protocol (MSRP), a nationwide initiative designed to assign patients with newly diagnosed myeloid cancers to molecularly targeted treatment trials. This protocol not only reflects the biological diversity of these diseases but also addresses the operational demands of multicenter precision oncology studies [9], [10].

To meet these challenges, a layered Fig.1 system architecture was designed to manage the entire lifecycle of precision trials, from patient enrollment through biomarker testing, trial matching, predictive analytics, and regulatory submission. Each layer interacts with the others through standardized data models, structured interfaces, and embedded clinical logic. This design ensures that information flows smoothly across trial components, enabling protocol compliance, real-time oversight, and regulatory alignment.

A. Data Aquisition and Harmonization

The first operational layer of the framework is dedicated to capturing and harmonizing data from multiple clinical entry points through the Medidata Rave electronic data capture system and the Medidata Patient Cloud [11]. Trial sites record structured information such as patient demographics, baseline laboratory values, bone marrow morphology, cytogenetic results, physical examination findings, and adverse events into protocol-specific case report forms. In parallel, patient-reported outcomes and electronic informed consent are collected through Patient Cloud, which synchronizes directly with the central trial database in real time.

Data integrity at this layer is maintained through embedded validation rules and live consistency checks. Each entry is time-stamped, source-tracked, and logged with a full audit trail to ensure compliance with 21 CFR Part 11 requirements and adherence to ICH-GCP standards for traceability [12]. The underlying data models are aligned with CDASH standards and preconfigured for seamless downstream conversion into SDTM datasets, supporting both clinical analysis and regulatory submission [14].

A defining capability of this layer is the integration of high-resolution biomarker data within clinically relevant timeframes. Bone marrow and peripheral blood specimens are routed to central molecular laboratories for rapid testing, with results typically available within three business days. These results encompass mutational profiling for FLT3, IDH1, IDH2, NPM1, and related markers, alongside karyotyping and disease classification outputs. All findings are coded as structured fields and tied to specific testing windows, ensuring temporal synchronization with downstream analytics and trial-matching logic [15].

B. Biomarker Interpretation and Protocol Rule Engine

The eligibility and stratification framework defined within the Master Screening and Reassessment Protocol (MSRP) is encoded into a dynamic rule engine that evaluates patient data in real time. The encoded rules include both static and temporal thresholds such as:

- ECOG performance status ≤ 2
- Bone marrow blast percentage $\geq 20\%$
- Absolute neutrophil count $\geq 0.5 \times 10^9/L$
- Platelet count $\geq 50 \times 10^9/L$
- Acceptable prior therapies within protocol-defined lookback periods
- Absence of major comorbidities or active secondary malignancies

As each data element is entered, the system automatically runs background checks against the rule set. Interactive dashboards then display eligibility status through visual flags, such as *"eligible," "pending value,"* or *"excluded."* Where clinically justified and permitted by protocol, manual overrides can be applied by the data review team to ensure patient safety and preserve trial flexibility.

A dedicated biomarker integration module further refines stratification by synthesizing molecular testing results. Detected mutations are cross-referenced with available subprotocols within the tiered structure of the MyeloMATCH system. For instance, a patient harboring an IDH2 mutation and meeting all other requirements may be assigned to a Tier 1 induction study investigating IDH2-targeted combination therapies. Patients who do not meet molecular assignment criteria are directed into the Tier Advancement Pathway (TAP), allowing continued participation in the trial and providing opportunities for reassessment as disease progression or protocol expansion creates new eligibility windows.

C. Patient Stratification and Trial Matching

Stratification in the system occurs at the intersection of molecular eligibility, clinical status, and the availability of active protocol windows. To operationalize this, a tiered logic structure is implemented that mirrors the master protocol design [16]:

- Tier 1 (Induction Studies): Newly diagnosed, untreated patients.
- Tier 2 (Refractory/Residual Disease): Patients with partial response or incomplete remission.
- Tier 3 (Post-Remission/Consolidation): Patients who achieve complete remission and transition to maintenance.
- Tier 4 (Minimal Residual Disease): Patients with no overt disease but detectable molecular persistence.

Eligibility across these tiers is monitored continuously through streaming data inputs. This allows reassessment and reallocation of patients as their disease evolves or as new trial arms become available. The result is a shift from a static, one-time screening event toward a longitudinal matching process that adapts dynamically to both the patient's clinical course and the evolving trial landscape [17].

Matching recommendations are generated by the decision engine and displayed in a clinical assignment dashboard. The dashboard provides full context for each assignment, including the molecular rationale, associated protocol codes, timing considerations, and links to supporting documentation such as biomarker laboratory reports and eligibility flag logs. In doing so, every matching decision remains evidence-based, auditable, and aligned with the most current trial configuration [18].

D. Predictive Analytics and Monitoring Intelligence

To enable proactive oversight and minimize risk, the system architecture integrates predictive machine learning modules trained on anonymized historical clinical trial datasets. These models are embedded within the analytics layer and updated continuously through rolling data snapshots, ensuring that predictions remain adaptive to evolving trial conditions [19]. The predictive framework delivers three core capabilities:

1. *Treatment Matching Probability Estimation:* Early screening data are analyzed to forecast the likelihood of a patient qualifying for an open subprotocol. This guides enrollment prioritization and informs the urgency of biomarker testing.
2. *Adverse Event Risk Modeling:* Time-sensitive algorithms evaluate pre-treatment laboratory results, disease burden, and pharmacodynamic indicators to identify patients at elevated risk for Grade 3 or 4 toxicities.
3. *Protocol Deviation Prediction:* Site-level metrics such as delayed visits, incomplete data entry, and consent-to-randomization lag are continuously monitored. Real-time models calculate deviation risk scores, enabling targeted interventions by monitoring teams [20].

Each prediction is paired with interpretability outputs, generated using SHAP-based explanations, so that trial

coordinators and site monitors can clearly understand the clinical variables driving each signal. Outputs are presented through interactive dashboards and seamlessly integrated into site audit workflows, promoting transparency and operational accountability [21].

E. Regulatory Documentation and Submission Engine

To reduce administrative workload and accelerate regulatory preparedness, the framework incorporates a document generation module powered by large language models (LLMs). The module is trained on publicly available investigational new drug (IND) submissions, clinical study summaries, and standardized filing templates from prior hematologic oncology trials [22].

Structured data streams from Medidata Rave are automatically transformed into preformatted regulatory content, including:

- Clinical Study Report sections (e.g., 2.7.1, 2.7.3)
- Adverse event narratives and listings
- Define.xml files and annotated case report forms (CRFs) for SDTM submission
- Protocol deviation justification reports

Each generated document undergoes automated validation for CDISC standards and FDA eCTD module alignment, ensuring readiness for regulatory submission. Outputs are stored with versioning metadata and linked to complete audit trails from study conduct, maintaining traceability and compliance while significantly reducing the manual burden of documentation.

This integrated architecture demonstrates a scalable, intelligent foundation for modern treatment trials in myeloid cancer. By combining biomarker resolution with protocol automation, machine learning foresight, and regulatory fluency, the system ensures not only optimal patient assignment, but also the operational excellence required for translational success in precision hematology.

III. Clinical Data Analysis and Methods

Precision-guided therapy in myeloid malignancies relies on a disciplined, transparent, and reproducible process that transforms raw biological and clinical measurements into validated, protocol-aligned decision frameworks. Within the MyeloMATCH infrastructure, the analytical rigor of treatment studies depends on the system's ability to extract, clean, structure, validate, and operationalize complex data streams spanning blood-derived biomarkers, bone marrow pathology, patient performance metrics, and genomic signatures. To further strengthen model robustness, contextual augmentation with open-source clinical–genomic resources, such as the Beat AML and TCGA-LAML datasets, is integrated into the workflow. These external datasets extend predictive modeling power while providing benchmark distributions that capture population-level diversity, ensuring that trial analyses remain both biologically grounded and generalizable [23], [24], [25].

A. Foundation of Data Capture and Continuity

All data within the framework originates at the site level, where structured electronic case report forms (eCRFs) in Medidata Rave capture core laboratory and diagnostic parameters. These include complete blood count (CBC) metrics such as absolute neutrophil count (ANC), hemoglobin (Hb), hematocrit (Hct), platelet count, and blast percentage as well as comprehensive metabolic panels evaluating renal and hepatic function through markers like creatinine, ALT, AST, and bilirubin. Bone marrow aspirates and biopsies are digitized and annotated for cellularity, fibrosis grading (MF-0 to MF-3), and blast infiltration, with morphological findings standardized across sites using established pathology lexicons [26], [27].

Central laboratory submissions extend the data pipeline to include advanced modalities such as next-generation sequencing panels (e.g., FoundationOne Heme), fluorescence in situ hybridization (FISH) for chromosomal translocations, and multiparametric flow cytometry for minimal residual disease (MRD) assessment [28].

To preserve temporal fidelity, all entries are time-stamped and aligned with predefined visit windows (e.g., Baseline, Day 1, Day 8, Day 15, End of Induction). Patient identifiers are indexed within a site–subject hierarchy, enabling precise linkage of biomarker emergence to specific intervention time points. This temporal structure provides the foundation for modeling treatment efficacy in a reproducible and clinically interpretable manner.

B. Quality Assurance and Rule based Preprocessing

Data integrity within the framework is safeguarded through a two-tiered validation schema. The first tier applies atomic-level constraints to individual variables, including permissible ranges for blast percentage (0–100), physiologically valid ranges for white blood cell counts, and unit normalization across contributing clinical laboratories worldwide. This level of validation ensures structural consistency and seamless interoperability with Medidata's backend systems.

The second tier enforces protocol-level logic. Eligibility rules are evaluated dynamically as patient data are entered, allowing immediate stratification against active treatment arms and Tier Advancement Pathway (TAP) options. For example, a 65-year-old patient with an ECOG performance score of 2, a marrow blast percentage of 25%, and an IDH2 mutation would be provisionally matched against available induction studies while simultaneously assessed for fallback TAP eligibility. Cross-variable inconsistencies such as a mismatch between cytogenetic risk classification and next-generation sequencing (NGS) results—trigger automated Medidata queries that require adjudication before the data are released into downstream modeling workflows.

To enhance robustness, the system incorporates open-source reference datasets. Benchmark envelopes derived from the Beat AML study guide anomaly detection, such as expected blast decline kinetics following induction therapy or median platelet recovery times. Similarly, the TCGA-LAML cohort provides population-scale priors on mutation frequencies, co-occurrence patterns, and variant allele frequency (VAF) distributions, ensuring that validation processes are grounded in real-world biological variation.

C. Data Cleaning, Missingness Resolution, and Feature Structuring

High-dimensional clinical data requires meticulous preprocessing to ensure analytical reliability. Outlier detection is applied early in the pipeline, with interquartile range (IQR) methods used for skewed distributions and

standard deviation filtering applied to normally distributed variables. Context-sensitive thresholds are enforced for instance, hemoglobin levels above 20 g/dL or absolute neutrophil counts (ANC) below 0.1×10^9/L are automatically flagged and reviewed against patient-specific treatment histories to confirm validity [29].

Missing values are resolved using class-specific imputation strategies. Continuous variables, such as creatinine, are imputed with k-nearest neighbor (KNN) algorithms guided by clinical proximity within multidimensional space. Categorical variables, including cytogenetic risk classifications, are addressed through frequency-weighted voting; in cases where imputation would compromise clinical logic, values are preserved as null [30].

Feature engineering transforms raw variables into predictive matrices optimized for downstream modeling. Time-dynamic features include blast reduction ratio (Day 15 versus Baseline), lag time from consent to first dose, ANC recovery duration, and probability of minimal residual disease (MRD) re-emergence. Cross-domain interaction terms, such as platelet-to-blast ratios or fibrosis-by-WBC multipliers, are also derived to highlight emergent risk phenotypes.

All engineered features are consolidated into structured data frames, each annotated with metadata tags that document imputation status, original data source, and transformation lineage. This annotation framework not only enables explainable machine learning but also provides the audit trails required for regulatory submission.

D. Biomarker Mapping and Eligibility Overlay

Molecular findings generated by central laboratories are encoded using HGVS nomenclature and annotated with pathogenicity references from databases such as ClinVar, COSMIC, and OncoKB. For example, a FLT3-ITD mutation with an allelic ratio greater than 0.5 is flagged as Tier 1 eligibility under the MYELOMATCH framework, whereas a concurrent TP53 mutation with high variant allele frequency (VAF) may trigger exclusion criteria or Tier Advancement Pathway (TAP) deferral. All eligibility logic is version-controlled and synchronized with ongoing study amendments through real-time rulesets, ensuring alignment with protocol updates [31], [32], [33].

To assess robustness, open-source datasets are leveraged to simulate counterfactual matching scenarios. For instance, applying MyeloMATCH eligibility logic to the TCGA-LAML dataset under varying VAF thresholds demonstrates how eligibility distributions shift across different demographic and molecular subgroups. These simulations provide an independent check on fairness, sensitivity, and treatment coverage within the stratification process [34].

Patients deemed ineligible for active arms due to incomplete biomarker panels or subthreshold mutation expression are systematically assessed for TAP eligibility. Automated flags are raised when clinical evolution or new assay availability indicates the need for reassessment. Biomarker dynamics such as rising IDH1 VAF or the emergence of FLT3-ITD at relapse trigger real-time rematching workflows, ensuring that trial allocation adapts continuously to the patient's evolving disease state.

E. Compliance Monitoring and Data Readiness for Modeling

Final datasets undergo rigorous validation for SDTM and ADaM conformance using a combination of open-source validation libraries and sponsor-defined macros. Each patient record is embedded with a complete transformation lineage, capturing source form identifiers, timestamp resolution, imputation status, and eligibility flags. Version-controlled data packages are then generated, containing Define.xml mappings, annotated case report forms (CRFs), and transformation logic preserved within traceable documentation containers [35], [36].

Readiness audits further enforce protocol compliance, data completeness, and temporal consistency. Only datasets that meet 100 percent of structural, logical, and regulatory validation criteria are advanced into secure machine learning pipelines. The framework's lineage-preserving features enable reverse mapping from model prediction outputs back to original clinical sources an essential capability for FDA auditability and for post-hoc reconciliation by clinical teams [37].

By unifying blood and marrow data streams with molecular, temporal, and contextual stratifiers, the MyeloMATCH clinical data framework establishes a replicable, regulatorily aligned, and biologically grounded platform. This foundation supports precision modeling, therapeutic reassignment, and ethically sound decision-making in clinical research for myeloid malignancies.

IV. Machine Learning Model Design

Designing machine learning models to guide treatment stratification in myeloid malignancies requires architecture that carefully balances predictive accuracy, interpretability, and clinical applicability. The curated dataset comprising blood-based parameters, bone marrow histopathology, and biomarker outputs from central laboratories is structured to support both supervised and semi-supervised learning approaches [38].

Predictive pipelines are developed to address multiple objectives: assessing patient eligibility, stratifying molecular risk, forecasting early treatment response, and anticipating adverse hematologic events. Each pipeline is constructed with built-in regulatory transparency, ensuring that output remains explainable and auditable. In addition, the models are fully integrated with Medidata's backend systems and designed with modular expandability, enabling seamless adaptation to future protocol amendments or new trial requirements [39].

A. Gradient Boosted Decision Trees (GBDT) for Eligibility Stratification

Gradient Boosted Decision Trees (GBDT) were selected for eligibility stratification because of their ability to capture non-linear interactions among heterogeneous clinical parameters. In this framework, eligibility modeling is framed as a multi-task classification problem, where each output node corresponds either to a specific trial arm or to a Tier Advancement Pathway (TAP) fallback option. Input features such as blast percentage, FLT3-ITD allelic ratio, marrow fibrosis grade, and cytogenetic risk include mixed data types and incomplete observations conditions that GBDT algorithms handle effectively.

From a mathematical perspective, the GBDT model constructs an ensemble of regression trees to iteratively minimize a differentiable loss function. The final prediction is expressed as:

$$f(x) = \sum_{m=1}^{M} \gamma_m h_m(x)$$

where $h_m(x)$ represents the output of the mmm-th decision tree, and γ_m denotes the corresponding weight optimized during training. The raw model outputs are subsequently passed through sigmoid activation functions to generate probability scores for each eligibility pathway. Thresholds are dynamically adjusted to remain consistent with protocol updates and eligibility amendments.

To ensure transparency, feature importance is first estimated using gain-based metrics and then refined with SHAP value analysis. This dual-layer interpretability allows attribution of predictions to specific input features, ensuring traceability and regulatory audit readiness while also supporting effective communication with trial sponsors.

B. Long Short-Term Memory Networks (LSTM) for Temporal Response Prediction

Long Short-Term Memory (LSTM) networks are employed to predict early treatment outcomes, such as complete remission (CR) or incomplete remission (CRi), by modeling time-series data from serial hematological assessments. The model inputs are longitudinal vectors derived from complete blood count (CBC) panels, including absolute neutrophil count (ANC), hemoglobin, and platelet counts measured at Days 1, 8, and 15. In addition, bone marrow blast percentages at Day 1 and Day 15 are incorporated to capture clearance kinetics, a critical predictor of treatment response.

The LSTM architecture leverages memory cells and gating mechanisms to retain long-range temporal dependencies while suppressing noise from irrelevant fluctuations. The governing equations for the cell state c_t and hidden state h_t are defined as:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) \\ i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i) \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh\,(W_c x_t + U_c h_{t-1} + b_c) \\ h_t &= o_t \odot \tanh\,(c_t) \end{aligned}$$

Here, σ represents the sigmoid activation function, and $\odot$ denotes element-wise multiplication. The output layer computes the probability of remission at Day 28 using a binary cross-entropy loss function. Training is performed with the Adam optimizer and incorporates learning rate decay to stabilize convergence. Model interpretability is achieved through SHAP value analysis across temporal dimensions, enabling the generation of patient-specific heatmaps that highlight the contribution of each variable at each time step. This visualization provides clinicians with actionable insights into the factors driving predicted treatment trajectories.

C. Regularized Logistic Regression for Adverse Event Forecasting

Accurately predicting high-grade adverse events, such as grade 3–4 neutropenia or thrombocytopenia, requires a modeling approach that is both robust and interpretable, particularly given the class imbalance typical of toxicity outcomes. Regularized logistic regression was selected for this task due to its transparency, stability under smaller sample sizes, and established acceptance in regulatory review processes.

Model inputs encompass baseline bone marrow characteristics, treatment regimen, and early-phase cytopenia kinetics. The logistic regression objective function, incorporating L2 regularization, is expressed as:

$$\min_{\theta} \sum_{i=1}^{n} \log\,(1 + e^{-y_i \theta^T x_i}) + \lambda \parallel \theta \parallel_2^2$$

where λ represents the regularization coefficient used to penalize overfitting, and θ denotes the model coefficients. Training is stratified by toxicity grade, and class weighting is applied to address imbalances between outcome categories.

The framework is extended to a multi-label classification setting using a one-vs-rest strategy, enabling simultaneous prediction of multiple adverse outcomes. To enhance interpretability, SHAP analysis is applied to map coefficients back to clinically meaningful input features such as the rate of ANC decline or the toxicity profile of a given treatment arm thereby supporting both clinical decision-making and regulatory auditability.

D. Semi-Supervised Learning with TCGA-LAML Integration

To expand model applicability for patients with incomplete laboratory results or partial biomarker profiles, a semi-supervised learning strategy is implemented. This approach employs a self-training loop in which a pre-trained GBDT model generates high-confidence predictions that are used as pseudo-labels for otherwise unlabeled data points, including those derived from TCGA-LAML [40].

The workflow iteratively augments the labeled dataset with these pseudo-labeled cases, followed by model recalibration at each iteration to enhance generalizability. The resulting model integrates both trial-collected records and harmonized public datasets, while maintaining strict adherence to eligibility logic and preserving full data provenance.

Each pseudo-label is annotated with confidence scores and flag indicators, ensuring that low-certainty imputations are excluded from influencing downstream analyses. This safeguards both data integrity and clinical validity, while enabling broader training coverage and improved robustness of the final predictive framework.

E. Interpretability and Deployment

All models incorporate SHAP-based interpretability to provide real-time clinician feedback on variable importance. These values are cached and linked to the original data lineage, facilitating regulatory traceability.

Each trained model is containerized using ONNX runtime, embedded within the Medidata clinical decision module. Deployment is subject to audit trails compliant with FDA 21 CFR Part 11, ensuring that all predictions can be traced back to their data source and transformation pipeline. also, architecture supports not only predictive accuracy but

also clinical accountability and regulatory defensibility [41].

V. ML Model Performance and Benchmarking

Evaluating machine learning models in a clinical trial environment, particularly in the context of precision-targeted interventions for myeloid malignancies—requires standards that go beyond conventional accuracy metrics. A robust evaluation framework must validate each model's reliability, clinical alignment, regulatory interpretability, and generalizability across heterogeneous patient populations. Within the MyeloMATCH trial architecture, model assessment was therefore conducted under a structured benchmarking protocol consistent with FDA guidance, ICH E6(R2), and Good Clinical Data Management Practices (GCDMP) [42], [43], [44].

The eligibility stratification engine, developed using Gradient Boosted Decision Trees (GBDT), was evaluated with stratified k-fold cross-validation (k=10k=10k=10). Stratification ensured balanced representation of eligibility-positive and eligibility-negative cases across folds, preserving class distribution and reducing sampling bias. Each fold underwent independent preprocessing, replicating a prospective deployment setting where data ingestion, cleaning, and transformation are performed separately. Model discrimination performance was quantified using the area under the receiver operating characteristic curve (AUROC). Across repeated validation cycles, the GBDT consistently achieved AUROC values between 0.915 and 0.938, with a mean calibration error of 0.034 as measured by the Brier score. Raw probability outputs were further calibrated using isotonic regression to align predicted eligibility scores with actual observed inclusion rates. A dynamic thresholding module was also integrated, allowing eligibility cutoffs to adapt seamlessly to protocol amendments and evolving regulatory criteria [45].

Beyond discrimination and calibration, explainability formed a critical dimension of model evaluation. SHAP (SHapley Additive exPlanations) values were computed for each prediction, with distributions aggregated across eligible and ineligible cohorts to identify global patterns of feature influence. These interpretability plots were routinely shared with clinical monitors and principal investigators for clinical plausibility review. For instance, in predictions involving FLT3-mutated subtypes, SHAP consistently assigned higher importance to blast percentage, FLT3-ITD allelic burden, and marrow fibrosis grade mirroring established clinical determinants of trial eligibility. This dual focus on quantitative performance and transparent interpretability ensures that the models are both scientifically rigorous and regulatorily defensible.

The temporal forecasting of early hematologic response, modeled through a Long Short-Term Memory (LSTM) neural network, required a validation strategy sensitive to sequential data dependencies. Since the input included CBC and bone marrow assessments at fixed intervals (Day 1, Day 8, Day 15), random sampling risked data leakage across folds. To address this, nested time-series cross-validation was employed, with patient sequences partitioned by enrollment period. The output variable binary response at Day 28 (CR/CRi) was predicted using a binary cross-entropy objective. The LSTM achieved a mean AUROC of 0.887 and a precision-recall AUC of 0.842 across folds. Importantly, recall for true responders remained above 90% in internal validations, ensuring no potentially responsive patients were excluded from trial consideration. To mitigate overfitting, the architecture incorporated dropout layers and temporal batch normalization. Temporal SHAP analysis was performed across time steps, generating clinician-facing heatmaps that highlighted how platelet recovery or marrow blast clearance influenced prediction confidence at each stage [46].

For adverse event forecasting particularly the onset of Grade 3–4 neutropenia or thrombocytopenia a regularized logistic regression model was employed due to its interpretability and robustness in small-sample settings. Model performance was assessed via stratified bootstrap sampling, generating 1,000 resampled datasets to estimate variance across AUROC, sensitivity, specificity, and F1-score. Paired t-tests were used to evaluate subgroup differences (e.g., treatment arm or marrow profile). Hyperparameters were optimized using grid search within a nested cross-validation framework, balancing model bias and variance. Final outputs were compared against expert clinical adjudications to confirm alignment with safety board reviews, ensuring predictions were clinically credible as well as statistically sound [47].

Model fairness and calibration were systematically evaluated. Performance was disaggregated by demographic and clinical subgroups, including age, gender, cytogenetic risk class, and ECOG performance status. Calibration drift was monitored over successive enrollment cohorts, with recalibration pipelines triggered when deviations emerged. Temperature scaling and Platt scaling were applied as corrective methods to sustain alignment between predicted and observed outcomes.

To improve generalizability in the presence of incomplete biomarker panels, a semi-supervised learning extension was introduced. Public datasets such as TCGA-LAML were integrated through a self-training loop, where a GBDT model generated high-confidence pseudo-labels. Samples meeting a probability threshold of greater than 0.95 were iteratively added to the training set. Domain adaptation error, measured by KL divergence between trial and TCGA prediction distributions, remained stable across iterations. Incorporation of pseudo-labeled data improved AUROC by 4.2% in patients with partial biomarker coverage, demonstrating the value of harmonizing trial and external datasets [48].

All benchmarking artifacts including performance logs, SHAP attribution maps, cross-validation outputs, and calibration charts were version-controlled and stored within the Medidata audit trail system. This repository, fully compliant with 21 CFR Part 11, provides end-to-end traceability from raw data ingestion to model deployment. Each model version is linked to its dataset hash and preprocessing pipeline, ensuring reproducibility, clinical accountability, and regulatory defensibility. Collectively, this evaluation framework ensures that ML-driven decision support in MyeloMATCH meets the dual standards of scientific rigor and regulatory readiness.

VI. Results and Discussion

The deployment of advanced machine learning algorithms within a clinically stratified, protocol-constrained, and biomarker-driven trial ecosystem such as

TABLE I
MODEL PERFORMANCE METRICES

Model	Use Case	Mean AUROC	Precision	Recall (Sensitivity)	Calibration Error (Brier)	SHAP Interpretability Agreement (κ)
GBDT	Eligibility Stratification	0.927	0.894	0.938	0.041	0.81
LSTM	Day 28 CR/CRi Response Forecast	0.892	0.872	0.913	0.036	0.78
Logistic Regression	Grade 3–4 AE Prediction (Neutropenia)	0.846	0.789	0.834	0.052	0.75
Semi-Supervised GBDT (TCGA Mix)	Cross-domain Eligibility Forecasting	0.895	0.854	0.902	0.049	0.74

MYELOMATCH requires an interpretation framework that is both analytically rigorous and clinically meaningful. Each modeling engine Gradient Boosted Decision Trees (GBDT), Long Short-Term Memory (LSTM) networks, and regularized logistic regression was evaluated against predefined baselines aligned with clinical trial data standards and regulatory audit requirements. This section TABLE II provides an integrated discussion of model outputs, performance variability across subpopulations, predictive accuracy, clinical relevance, and comparative benchmarking across key endpoints.

The GBDT-based eligibility stratification model demonstrated strong discriminatory capacity in assigning patients to protocol-eligible treatment arms. Using 10-fold stratified cross-validation, the model achieved an average AUROC of 0.927 (95% CI: 0.914–0.942). Sensitivity for high-risk FLT3-ITD positive subtypes reached 93.8%, while specificity for IDH1/2-negative cohorts remained stable at 88.4%. Calibration was evaluated using Brier scores (0.041) and isotonic regression, with results closely aligned to observed enrollment outcomes. Importantly, SHAP-based feature interpretation reinforced clinical plausibility: marrow blast count, cytogenetic risk classification, and platelet count consistently ranked among the top contributors in more than 80% of cases. Incorporating SHAP interpretability not only preserved performance but also improved clinical adoption, as confirmed by strong agreement between model explanations and investigator assessments ($\kappa = 0.81$).

The LSTM model for temporal response prediction yielded robust performance despite heterogeneity in sequential hematologic data collected on Days 1, 8, and 15. The model achieved a mean AUROC of 0.892 for predicting Day 28 CR/CRi outcomes, with sensitivity reaching 91.3% among patients achieving full marrow recovery by Day 15. Notably, the model captured non-linear temporal dynamics: patients with delayed blast clearance between Day 15 and Day 28 showed probability shifts from below 0.5 to above 0.8, validating the LSTM's capacity to model recovery trajectories without overfitting. Temporal SHAP visualizations provided further interpretability, demonstrating shifting feature contributions across time windows for instance, platelet counts were minimally influential at Day 1 but rose in importance by Day 15, consistent with known hematopoietic rebound physiology. Cross-validation folds were aligned with trial accrual periods to mitigate cohort drift and preserve deployment fidelity.

The logistic regression–based adverse event forecaster was designed to identify early-onset Grade 3–4 hematologic toxicities, which represent critical safety endpoints in myeloid cancer trials. The model achieved moderate to high accuracy, with AUROC values ranging from 0.821 to 0.846 across neutropenia and thrombocytopenia predictions. Incorporating L2 regularization effectively minimized overfitting, particularly for sparse outcomes such as late-onset Grade 4 cytopenias. Precision-recall analyses further confirmed robustness, with average precision scores of 0.789 for neutropenia and 0.765 for thrombocytopenia. SHAP interpretability highlighted clinically consistent predictors, including pre-treatment marrow hypocellularity, the slope of ANC decline, and baseline ECOG status. Model outputs were cross-validated against manual toxicity grading logs from the Data Safety Monitoring Board (DSMB), achieving a concordance rate of 92.4% with adjudicated outcomes. This high alignment not only reinforced clinical reliability but also satisfied FDA traceability and safety reporting requirements under 21 CFR Part 312 [49].

The semi-supervised learning module expanded model coverage for patients with incomplete biomarker panels or missing laboratory assessments by incorporating TCGA-LAML data. Pseudo-labeling was restricted to predictions exceeding 95% confidence, while low-certainty cases (<90%) were excluded from augmentation cycles. After three iterations of self-training, the retrained model improved its macro-F1 score by 4.2%, with AUROC gains of 3.8% for eligibility prediction and 5.1% for early response forecasting. Domain adaptation performance, measured through KL divergence between trial and pseudo-labeled TCGA distributions, improved from 0.231 to 0.089, indicating successful generalization with minimal drift. Validation on a held-out trial cohort confirmed calibration fidelity, meeting regulatory expectations for external dataset integration.

All models were benchmarked not only on statistical accuracy but also on deployment-readiness. Evaluation criteria included interpretability (quantified by a SHAP transparency index), compatibility with Medidata's ONNX container runtime for deployment, and audit trace completeness. Comprehensive audit logs recorded model parameters, transformation pipelines, imputation methods,

and lineage mappings that traced each prediction back to its originating case report form (CRF) field and timestamp. Internal QA simulations of FDA 21 CFR Part 11 inspections were conducted, with all models passing without non-conformities.

A comparative summary TABLE III of evaluation metrics across eligibility, response, toxicity, and generalization tasks contextualizes performance across the entire modeling suite, demonstrating both clinical robustness and regulatory defensibility.

This comparative analysis demonstrates that the modeling framework functions not as a collection of isolated statistical tools but as an integrated, audit-traceable, and clinically coherent decision support system. Model outputs are seamlessly embedded into the trial's eligibility adjudication interfaces and toxicity surveillance dashboards, augmented with traceability flags, SHAP-based interpretability overlays, and version control hashes. These features ensure reproducibility, regulatory accountability, and ethical deployment, establishing a reliable foundation for precision-guided decision-making in myeloid oncology.

VII. Conclusion

This work represents a paradigm shift in the conduct of clinical research for myeloid malignancies, enabled by the convergence of real-time data capture, advanced machine learning models, and regulatory-aligned infrastructure. At its foundation, the MYELOMATCH AI-driven framework demonstrates how structured oncology trial data spanning electronic case report forms, molecular diagnostics, and longitudinal laboratory series can be transformed into intelligent, reproducible clinical decision support.

The system integrates a modular suite of machine learning approaches: Gradient-Boosted Decision Trees (GBDT) for eligibility stratification, Long Short-Term Memory (LSTM) networks for response prediction, and regularized logistic regression for adverse event forecasting. Together, these models extend beyond static classification, delivering temporally aware, contextually grounded predictions. Containerized in ONNX and deployed within Medidata's Clinical Cloud, the models are fully linked to annotated CRFs and embedded in audit-logged workflows, ensuring explainability, clinical traceability, and compliance with FDA 21 CFR Part 11 [50].

The underlying data engineering pipeline provides dynamic intake, quality assurance, harmonization, and feature engineering, ensuring that predictive outputs are both biologically valid and regulatorily defensible. SHAP-based interpretability overlays offer clinicians and regulators human-readable transparency for each prediction, reinforcing trust and enabling real-time validation of algorithmic decisions.

Importantly, the platform is designed not only for retrospective analysis but also for prospective application. Its architecture supports dynamic reassessment, adaptive treatment assignment, and near real-time safety monitoring. Semi-supervised learning pathways extend model applicability to patients with incomplete biomarker panels, bridging real-world data gaps and broadening the impact of precision oncology.

Looking ahead, the framework provides a foundation for adaptive trials and federated AI systems capable of continual learning across distributed networks. As additional datasets such as BeatAML and TCGA-LAML are integrated, external validation will enhance generalizability and promote cross-institutional standardization. Moreover, alignment with emerging Medidata modules such as Rave RTSM and Patient Cloud ePRO opens new avenues for embedding predictive analytics across the entire patient journey from screening and enrollment through treatment and long-term follow-up.

In summary, this work demonstrates a scalable, ethically grounded, and regulatorily aligned realization of AI-enabled clinical trial design. It illustrates how modern data science when carefully architected and deeply embedded within clinical workflows can improve trial efficiency, enhance patient safety, and strengthen scientific rigor. The system not only aligns with the future of decentralized, digital-first clinical research but actively helps define it.

VIII. References

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